ATRIA

Dynamic Emergency Department Triage & Reassessment
Implementation Explained

What has been built, how each piece works, and what technology it runs on. Written to be read by a triage nurse, a clinical governance reviewer and an engineer — in that order of priority.

Document	Implementation companion to the ATRIA PRD v1.0
Status	Working prototype. Clinical validation required.
Team	Digital Ninja · IIT Kanpur
Challenge	Accenture Innovation Challenge 2026 · Round 2, Track 2
Code	~5,800 lines Python · ~2,600 lines TypeScript · 254 automated tests
Safety status	No threshold here has been approved by a clinical governance body. Not for use on real patients.

Document map

- **1–2** — the problem, and what ATRIA is
- **3–6** — how it works: the four layers, the blind workflow, refusing to answer, the record
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- **11–13** — what we measured, what we found by building, honest limits
- **14** — running it, and a demo script

1. The problem

Triage happens once, at the door. Patients deteriorate afterwards.

When someone arrives, a nurse assigns a priority from 1 (resuscitate now) to 5 (can safely wait). That number then follows them into the waiting room and mostly stays there. The patient's condition does not. Someone who looked stable at 9:00 can be in trouble by 9:40, and in a busy department nothing is systematically watching for it.

Two failures follow. People deteriorate unseen in the waiting room. And the queue drifts towards first-come-first-served, which is not the same as sickest-first.

2. What ATRIA is

A continuously re-ranking attention list. It watches vital signs as they are re-taken and changes the order in which patients should be seen. It supports the nurse. It never diagnoses, never suggests treatment, and can never make a patient less urgent on its own.

> **In one sentence.** ATRIA turns triage from a decision made once into a decision kept under review — while leaving every actual decision with a human, and writing down who made it.

3. How it works — the four layers

The system has four layers. The most important thing about them is not what they do, but **which are allowed to decide anything.**

Layer	What it does	Authority
0 — Safety rules	Eleven fixed rules taken from published clinical guidance. Oxygen below 90%. Blood pressure below the minimum *for that patient's age*. Active seizure. Stroke signs inside the treatment window. Runs first, every time, with no model involved.	**Decides.** If a rule fires the patient goes to priority 1 and no model output can overrule it.

1 — Acuity scorer	The only trained model in the system. Reads 23 pieces of information available within five minutes of arrival and produces a suggested priority plus a confidence.	*Suggests only.*
2 — Trajectory watcher	Compares each new set of readings with the previous ones. Oxygen down 3 points, blood pressure down 15, heart rate up 20, breathing rate up 6 — the direction of travel rather than the snapshot.	*Re-orders the queue.*
3 — The human	The nurse's workflow, and a tamper-evident record of every decision.	**Decides.** Every reduction in urgency is a human act.

> **Only Layer 1 contains a trained model.** Layers 0, 2 and 3 are ordinary code — fixed numbers, arithmetic and a checklist. That was a deliberate choice. The parts that can push a patient to the front, refuse to answer, or record what a clinician did should be readable line by line by someone who does not trust machine learning at all.

The ratchet

Layers 0, 1 and 2 can all *raise* urgency. **None of them can lower it.** Only a clinician can, and it is recorded with their name, their reason, the exact version of the model they disagreed with, and the readings it saw.

Waiting is not the same as getting sicker

An early version treated time spent waiting as deterioration, which inflated the whole board. Now, being overdue for a re-check *flags* a patient for a re-look but does not raise their priority. Only at three times the safe waiting interval does it escalate by one, as a backstop.

4. The idea the whole design rests on

The nurse decides first. ATRIA speaks second.

When a nurse opens a patient, ATRIA's suggestion is not on the screen. It is not hidden with styling either — **it is not sent to the browser at all.** The nurse enters their own priority. Only then does the server release the suggestion, together with a one-time ticket proving the nurse's answer was stored first. Ask early and the server refuses outright.

- **Anchoring.** Show a clinician a number first and they drift towards it. That is not carelessness, it is how attention works under pressure. Hiding it until the human commits is the cheapest known defence.
- **It creates evidence.** Because the nurse always answers blind, every patient produces an independent human judgement — the only way to ever know whether the machine adds anything.

For the same reason the model is forbidden from reading the nurse's priority as an input. A model told the answer cannot meaningfully disagree with it.

> **We measured what that costs.** Letting the model read the nurse's priority would lift its accuracy score from **0.809** to **0.859**. We gave up that 0.05 deliberately, and we put it on the slide rather than hiding it.

What happens after the reveal

The nurse chose	What happens
The same as ATRIA	Confirm and move on.
More urgent than ATRIA	Straight through, no explanation asked for. A clinician raising urgency is never questioned.
Less urgent than ATRIA	Blocked until they say why. This is the direction that can cause harm.
Anything, but a safety rule fired	Blocked until justified, and the charge nurse is notified.
Anything, and ATRIA would not score	Blocked until justified.

If the patient's condition changes, reporting it clears the sign-off and starts a fresh blind cycle. The previous suggestion is discarded rather than shown again — carrying it over would anchor the very decision that has to stay independent.

5. When the system refuses to answer

A triage system that always produces a number is lying some of the time. ATRIA has two ways of saying it does not know.

- **Too little to go on.** Fewer than three vital signs recorded and no score is produced at all. You cannot triage a sentence.
- **Genuinely ambiguous.** When the signs point at two different failing body systems at once, it says so rather than picking one.

Neither is treated as reassuring. A patient nobody can classify goes *ahead* of everyone already cleared, and both cases go into the record.

> **Two different kinds of confidence.** ATRIA reports how sure it is about *how urgent* someone is separately from how sure it is about *what is going wrong*. A cold, badly injured patient scores high urgency and low cause-confidence: we are certain they are critical and honest that we cannot say which system is failing. That patient needs a doctor now, not a better score.

6. The record

Every event — arrival, the nurse's answer, the reveal, the sign-off, an override — is written into a chain where **each entry seals the one before it**. Change any past record and the seal no longer matches, so tampering is visible rather than silent. Corrections are added as new linked entries; nothing is ever rewritten.

The chain is stored on disk and rebuilt when the system restarts, because a tamper-evident record that vanishes on restart proves nothing. The database itself refuses edits and deletions, so tampering fails when it is attempted rather than being caught later by a check nobody ran.

7. What a nurse actually does

Three screens, one patient at a time.

> **The short version.** Tap a patient. Enter your priority. Look at the comparison. Sign off. If you disagree by making someone *more* urgent, nothing stops you and nothing asks why.

The board

A list of everyone waiting, most urgent at the top, re-ordering itself as readings arrive. When someone moves up, the row **slides** to its new place so you can see it happen rather than noticing a different order. Each row shows their ticket, priority, which area they belong in, how long they have waited, and why ATRIA has them where it does.

Counters across the top: how many waiting, how many in treatment, how many have been moved up, and how many ATRIA *would not score* and has handed to a human.

Using it without a mouse

Key	Does
`j` / `k`	Next / previous patient
`1` – `5`	Record your priority for the selected patient
`Enter`	Confirm the step you are on

If you use a screen reader, patients moving up are announced out loud — an animation is not information everyone can use.

Step 1 — your decision

Five large buttons, 1 to 5, each with a plain description. **There is nothing from ATRIA on this screen.** You are not being tested and there is no right answer to guess at.

Step 2 — compare

Your number and ATRIA's, side by side, with a plain-English banner saying what the difference means and whether anything is needed from you.

Step 3 — sign off

If a reason is required you pick one from a list. If you pick **"something else"**, you are asked to write what you saw — because "something else" names no reason at all, and a record that says "other" tells a reviewer nothing.

The sign-off is saved with your name, the time, the readings it was based on, and the version of the model you were comparing against.

> **Things that will not happen to you.** A patient will not be taken through to treatment while you are part-way through assessing them — they are held from the moment you enter your priority until you sign off. The record panel will not switch to a different patient underneath you. And you will never be asked to justify making someone *more* urgent.

8. The technology

Nothing exotic. Every choice below was made so that a hospital IT department could look at it without surprises.

Part	Built with	Why
Clinical engine the four layers	Python 3.12, pandas, NumPy	Plain code with no framework around it, so the safety logic can be read on its own. It has no idea what is drawing it.
The model	scikit-learn — gradient-boosted decision trees	Trees rather than a neural network: fast, works with missing values, and each prediction can be explained by the features that drove it.
Safety rules	A YAML file of thresholds, each with a clinical citation	A clinician can read and change the rules without touching code — which is what clinical governance will require.
API and live feed	FastAPI, uvicorn, WebSocket — 23 endpoints	The board updates when something happens rather than polling on a timer.
Nurse board	Next.js 16, React 19, TypeScript, Tailwind CSS 4	The interface the PRD specifies. TypeScript catches a mismatch between screen and server at build time rather than in front of a patient.
Long queues	TanStack Virtual	Two hundred patients render smoothly because only the visible rows exist on the screen at any moment.
Charts	Recharts	The department flow forecast.
Second board	Streamlit	A simpler board over the same engine — proof the clinical logic is genuinely independent of any one interface.
The record	SQLite	One portable file an auditor can take away and check. No database server to install beside it.
Sign-in	JWT tokens, PBKDF2 password hashing	Standard, boring, and no password is ever stored in a readable form.
Hospital records	FHIR R4 over HTTPS, read-only	The international standard for exchanging health records.

Packaging	Docker Compose	The whole stack starts with one command.
Checks	pytest — 254 automated tests	Every safety rule above has a test that fails if someone breaks it.

9. How each piece is actually built

Layer 0 — the safety rules

A table of thresholds in a plain text file, each row carrying its clinical source. The code reads the table; it does not contain the numbers. Thresholds that depend on age — blood pressure and breathing rate — are looked up by age band rather than assumed adult.

The gate returns **three** answers, not two: the rule fired on a reading someone actually took; the rule only fired because a missing reading was assumed to be the worst case (priority 2, plus an instruction to go and measure it); or this data source has no such reading at all, so the rule cannot be judged. The third matters — treating a never-recorded value as an emergency would flag every patient in the department.

Layer 1 — the model

Trained on **560,486 real emergency visits** from three American hospitals. The data is split three ways: one part to learn from, one to calibrate its confidence, one held back entirely to measure it. The priority bands come from the distribution of risk across real patients, not from an arbitrary cut-off.

The nurse's own priority and the patient's race are excluded from the inputs — the first so the model can genuinely disagree with the nurse, the second because a model must not use race as a shortcut.

Confidence is calculated *per priority level* rather than on average. An average guarantee is met by being reliable about the common cases and unreliable about the rare ones — and the rare ones here are the critical patients.

Layer 2 — the trajectory watcher

Arithmetic on the last few sets of readings. Falls and rises beyond fixed amounts, plus shock index (heart rate divided by blood pressure) crossing 0.9. Re-checks are due at 5, 15, 45, 90 and 180 minutes for priorities 1 to 5.

Priorities are strictly banded. Waiting time and department pressure re-order patients *within* a priority and can never move one across a boundary. A maximally boosted priority 3 still sits below an untouched priority 2, and a test proves it.

Layer 3 — workflow and record

A small state machine: awaiting the nurse, compared, signed. The suggestion is added to the response only after the nurse's answer is stored, and the one-time ticket is spent on use — it was reusable at one point, which made it a password rather than a proof, and that was found by running thousands of simulated shifts rather than by a targeted test.

Sign-in and roles

Role	Can
Triage nurse	See the queue, assess, sign off, report a change, check patients in
Charge nurse	The above, plus acknowledgements, department view and history
Clinician	The above, plus **lowering a priority**
Flow coordinator	The department view only
Clinical governance	The decision history, read-only
Administrator	Everything, including switching shadow mode

Two rules worth stating out loud. **The person auditing the record cannot add to it.** And **the person managing bed pressure cannot lower a patient's urgency** to relieve it — downgrading is a clinical act.

Who you are comes from your sign-in, never from something the screen sends. A record you could sign with any name you liked would not be evidence of anything.

Shadow mode

Every layer runs and **nothing moves**. The department carries on exactly as before while ATRIA writes down what it would have done. The report says how often it would have escalated someone the current process left waiting — and ends with the specific patients to chart-review, because a percentage does not tell anyone whether ATRIA was *right* to disagree.

One exception: a Layer 0 safety rule still acts. Those are cited thresholds any triage protocol already applies, and withholding one would mean deliberately not acting on a measured oxygen level of 84% in order to keep an experiment clean.

10. What is implemented

Capability	Status	Note
Four clinical layers	BUILT	All rules cited and tested
Blind nurse-first assessment	BUILT	Enforced by the server, not the screen
Tamper-evident record, survives restart	BUILT	SQLite, edits and deletes refused
Sign-in and six roles	BUILT	Every endpoint checked by an automated test
Frozen, versioned model	BUILT	Stamped on every sign-off
Shadow mode	BUILT	With a review report
Nurse board (Next.js)	BUILT	Queue, assessment, department view, history
Degraded mode	BUILT	Model off, safety rules keep gating

Reading vitals from a hospital system	SANDBOX	FHIR R4, tested against a public test server. No hospital connected.
Writing back to a hospital record	BY DESIGN, NO	There is no code that could
Hospital single sign-on	NOT BUILT	Demo accounts only; a pilot needs real identity
Clinical governance approval	NOT STARTED	Every threshold is a default, not an approved value

11. What we measured

Every number below is produced by a script in the repository and regenerated by one command. None is typed into a slide by hand.

What	Result
Model accuracy	**0.809** on 560,486 real emergency visits. The published benchmark on the same data is 0.87 — using the nurse's priority and race, which we exclude.
Operating point	**95%** of seriously ill patients caught, matching the American College of Surgeons standard of missing no more than 5%. Tuned to that, not to looking accurate.
Trajectory watcher	Flags **32.2%** of patients later admitted against **12.2%** sent home — a 20-point difference that holds up statistically. Median warning: **164 minutes**.
Fairness	Gap between best- and worst-served groups reduced from **5.3%** to **2.1%**, measured on patients the correction was not fitted to.
Speed	**52 milliseconds** at the 95th percentile under three times the normal load.
Reliability	25 simulated shifts, 434 patients treated, 565 full nurse workflows, **no failures**.

12. Three things we only found by building it

None of these was in any plan. Each is a way the system could have quietly harmed someone.

The dataset that would have taught it the wrong lesson

In one hospital dataset, **every single one of the most critical patients had no vital signs recorded** — because the sickest bypass the paperwork and go straight to resuscitation. A model trained on it would have learned that an empty form means a dying patient: superb scores, and useless in any hospital with different paperwork. We excluded that dataset from training.

The model thought missing readings were good news

We blanked out each vital sign in turn and watched the score. A missing heart rate, breathing rate or blood pressure made the model rate the patient *safer*. In practice that means the patient nobody has got round to measuring is the one quietly pushed down the list. Scores are now held so that an unrecorded reading can never beat an average one.

Our own safety rules were wrong about children

The rules raised a low-blood-pressure alarm on a three-year-old at 88 — entirely normal at that age. The safety layer built to protect people was itself adult-calibrated. It now uses paediatric age bands.

13. Being honest about the limits

These are on the slide on purpose. A prototype that claims no weaknesses has not been examined carefully enough.

- **We measure "was admitted to hospital", not "died or went to intensive care".** Admission is a rough stand-in for how sick someone is. We checked both available datasets; neither records intensive-care timings.
- **We tried a sharper measure and it did not work.** Using a time-critical diagnosis instead of admission gave **+5.1 points with a range spanning zero**, on 19 patients. That is a negative result and we publish it — and it is the strongest argument we have for access to proper hospital data.
- **The trajectory watcher is validated on 159 real patient journeys.** A small number, and every figure now carries a range showing how uncertain it is.
- **The fairness gap is small, not zero.** Four patient groups are too small to measure reliably at all; they are named in the report rather than quietly dropped.
- **The three failure pathways are the classical triad, assumed.** Round 1 material names only one of them explicitly.
- **Nothing has been through clinical governance.** Every threshold is a sensible default, not an approved one.

14. Running it

Command	What you get
`make demo`	The engine and built-in board, port 8000
`make web`	The nurse board, port 3000
`make streamlit`	The second board, port 8501
`make shadow`	Everything running, nothing acting
`make scenarios`	Seven fixed demo cases, identical every run
`make freeze`	Retrain and pin the model
`make report`	Regenerate every measured number
`make test`	All 254 checks

Sign in as `nurse.demo` / `nurse.demo`, or `admin.demo` for everything. Demo accounts only — they do not exist in a real installation.

A four-minute demonstration

1. Open a patient. Point out there is **nothing from ATRIA on screen** — not hidden, not sent.
2. Enter a priority. The comparison appears.
3. Choose a *less* urgent priority than ATRIA — sign-off is blocked until you explain. Choose *more* urgent and it goes straight through.
4. Switch the model off. The safety rules keep working and keep flagging critical patients.

> **The closing line.** ATRIA is not a system that decides who is sick. It is a system that makes sure nobody is forgotten while they wait, and that every decision about them has a name attached to it.